

EXPERIMENT 1: Preparation of a Coordination Compound

Objectives

1. to synthesize a coordination compound
2. to examine some of the properties of transition metal ions

Introduction

Coordination compounds result when a Lewis acid and a Lewis base react together to form a Lewis acid-base adduct (or coordination complex). A classic example of this is the reaction of the electron deficient molecule, BF_3 (Lewis acid), with NH_3 (Lewis base) to form $\text{BF}_3 \text{NH}_3$ (Figure 1), which contains a coordinate covalent bond between the boron and nitrogen atoms, and in which the boron has an octet of electrons. The NH_3 molecule is called a *ligand* because of the fact that it has a lone pair of electrons which it can donate to the electron deficient BF_3 .

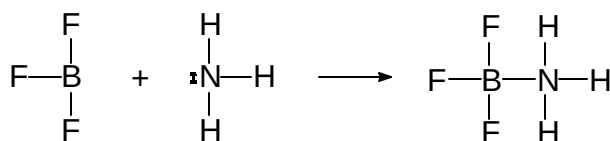


Figure 1

A similar situation occurs with transition metal ions, and this forms the basis for the vast and complex chemistry of those biological molecules which have such elements at their active sites (e.g. deoxy-myoglobin with an Fe^{2+} in the centre of a porphyrin ring with four nitrogen lone-pairs bonded to the iron centre).

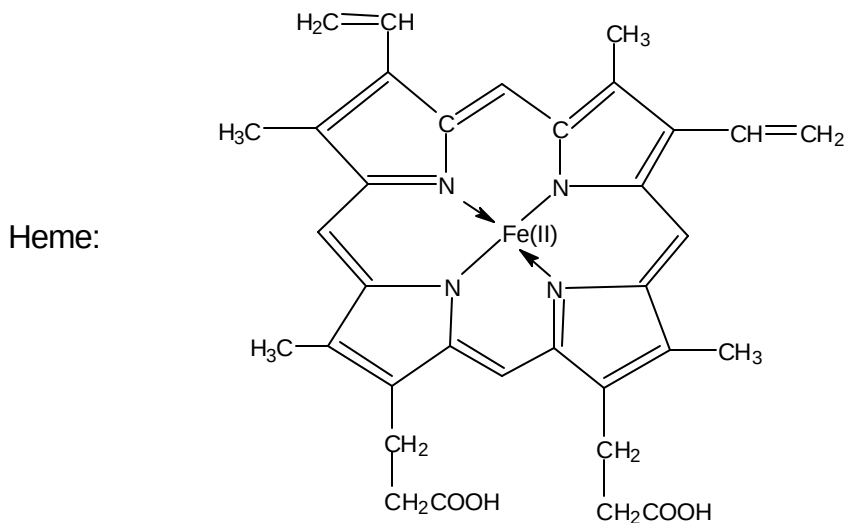


Figure 2

Transition metal salts found on the laboratory shelf are good examples of coordination complexes. For example, a bottle labelled $\text{NiCl}_2 \cdot 6\text{H}_2\text{O}$ should really be labelled as $[\text{Ni}(\text{H}_2\text{O})_6]\text{Cl}_2$, because the six water molecules are actually ligands which coordinate to the nickel centre to form an octahedral complex ion (Figure 3). Note that the **water molecules are all equivalent** and are positioned at the apices of an octahedron, with angles of 90° between adjacent $\text{Ni}-\text{OH}_2$ bonds. Water is called a monodentate ligand (one donor centre), but if more than one donor site is joined by some chemical linkage then polydentate ligands with several points of attachment result.

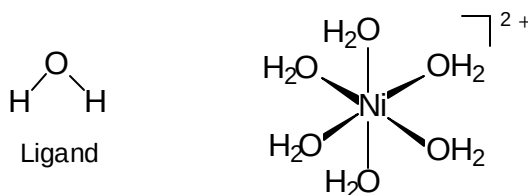
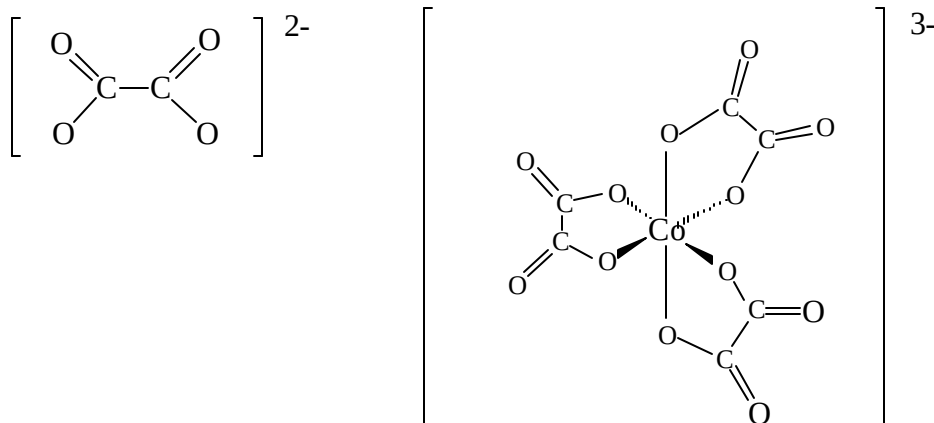


Figure 3

Polydentate ligands with six (i.e. hexadentate) or even more donor centres are known. The binding of polydentate ligands to a metal ion results in *chelation*, with the formation of stable *chelate* rings. The oxalate anion is a good *chelating* ligand, with two oxygen donor centres, and a double negative charge. These two factors impart high stability to the coordination complexes formed by oxalate.

The tris-oxalatocobaltate(III) ion is shown in Figure 4. This ion has the same octahedral geometry and oxidation number as the tris-oxalatoferrate(III) ion which you will be preparing in this experiment.



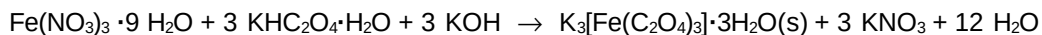
oxalate ion

tris-oxalotocobaltate(III) ion

Figure 4

CAUTION: Oxalic acid is a toxic compound, which can be absorbed through the skin. If skin contact is made, wash the affected area with lots of water. **Ethanol and acetone are flammable** and so all flames should be extinguished while carrying out filtration and washing steps during the following synthesis.

Note* The prelab exercise must be completed before you come to the lab.

Part I (a) Preparation of $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$ (Potassium tris-oxalatoferrate(III) trihydrate.)

Place a clean, dry 250 mL beaker on the top-loading balance, record its mass and weigh by difference about 8.1 g (record actual mass used) of iron(III) nitrate nonahydrate, $Fe(NO_3)_3 \cdot 9H_2O$, into it. Add 20 mL of deionized water and a few boiling granules to the beaker and heat it gently using a Bunsen burner. **Do not allow the solution to boil.**

Weigh by difference about 7.7 g (record actual mass used) of potassium hydrogen oxalate monohydrate, $KHC_2O_4 \cdot H_2O$ into a clean, dry weighing vial and add it, with stirring, to the beaker when all the $Fe(NO_3)_3 \cdot 9H_2O$ has dissolved.

Heat the mixture gently until all the solid dissolves, then, stirring carefully, add 10 mL of 6.0 mol L⁻¹ KOH. Stir the mixture until all of the brown precipitate which forms initially has dissolved. If the brown solid fails to dissolve you have made an error in your weighings and must begin again.

Filter the hot solution through a glass funnel fitted with a fluted filter paper into a 250 mL conical flask.

Slowly, with swirling, add 2 mL of 95% ethanol to the filtrate. Allow the solution to cool without disturbing it for at least 30 minutes to induce crystal growth. If the crystals have not appeared at this time add 3 more mL of 95% ethanol. Shake the flask vigorously and then allow it to stand undisturbed for a further 10 minutes to complete the crystallization.

When crystallization is complete, collect the product by suction filtration in a Buchner funnel. Wash the crystals twice with 10 mL portions of acetone. Break the suction before each addition by pulling the hose off the aspirator or loosening the funnel. **Warning: acetone and ethanol are flammable solvents. All flames must be extinguished in the lab before removing these solvents from the fumehood.**

Allow the crystals to dry for 5 minutes under suction. Then spread them out on a clean piece of smooth paper to air dry while you prepare the sample vial and label. Transfer the crystals to a pre-weighed vial and weigh the vial plus crystals. Record the yield, product formula and your name and bench # on the product label. Immediately wrap the vial in aluminum foil to protect your product from exposure to light and leave the product at your bench to be collected along with your report.

Part II Colours Associated with Ligand Changes in some Co-ordination Compounds of Copper(II).

Reactions of Copper(II)

Many transition metal ions are coloured because the energies required for the electronic transitions within their partially-filled d-subshells lie in the visible region of the electromagnetic spectrum. That is, visible light passing through their crystals or solutions is sufficiently energetic to raise a d-electron from the ground state to a higher energy level within the d-subshell. Light passing through the crystal or solution will have certain wavelengths absorbed. The colour you observe will be composed of the remaining visible wavelengths. For example, octahedral copper(II) complexes such as $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}$ are typically light blue. This means that the blue and possibly some adjacent green and violet wavelengths pass through the crystal while the lower energy red and yellow wavelengths are absorbed. The wavelength of light absorbed by the crystal and hence its colour will vary both with co-ordination number and ligand type, resulting in a change in colour of the solution when a new ligand is added.

The copper(II) ion readily forms coordination complexes with a variety of coordination numbers and geometries. These include four coordinate, square-planar complexes and five- and six-coordinate derivatives of the sp^3d^2 hybridized octahedral structure. In this experiment a solution of the six-coordinate complex ion $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ will be converted to the four-coordinate complex ion $[\text{Cu}(\text{NH}_3)_4]^{2+}(\text{aq})$ using aqueous ammonia then to the four-coordinate complex ion $[\text{Cu}(\text{Cl})_4]^{2-}(\text{aq})$ ion using aqueous HCl.

Procedure

1. Record the colors of the aqueous transition metal salts provided (see Table 3).
2. Take 1 mL of the $[\text{Cu}(\text{H}_2\text{O})_6]^{2+}(\text{aq})$ solution in a large test tube. Add one drop of 6 mol L^{-1} ammonia solution and note any change. Continue addition of the ammonia dropwise noting further changes, until any precipitate that forms finally dissolves. Record your observations.
3. To the same solution add 6 mol L^{-1} HCl dropwise, mixing at each addition, until no further changes occur. Record your observations in Table 4.

Laboratory Report	Date:	Lab. Slot:
	Name:	
Bench Number:		

Part I**Table 1**

Mass of $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{O}$ + vial + lid	_____
Mass emptied vial + lid	_____
Mass of $\text{KHC}_2\text{O}_4 \cdot \text{H}_2\text{O}$	_____

Table 2

Mass of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3 \text{H}_2\text{O}$ + vial + lid	_____
Mass vial + lid	_____
Mass of $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3 \text{H}_2\text{O}$	_____

Calculations:

Show calculations in detail with full identification of the terms, paying attention to units and significant figures.

1. Calculate molar masses and the number of moles for all reagents and your product.
2. Using the balanced equation for the reaction you carried out, calculate the limiting reagent.

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4. Calculate the theoretical yield of your product, $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$.

5. Calculate the percentage yield of the product.

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Part II**Reactions of copper****Table 3**

	Solution colour
Fe ²⁺	
Co ²⁺	
Ni ²⁺	
Cu ²⁺	
Zn ²⁺	

Table 4

[Cu(H ₂ O) ₆] ²⁺ (aq)	1 drop NH ₃ (aq)	Excess NH ₃ (aq)	Excess HCl(aq)
observation			

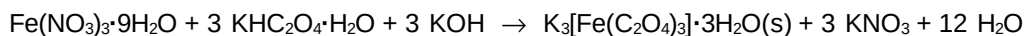
Question

1. Write net ionic equations to represent the reactions you have observed.

Prelaboratory Exercise	Date:	Lab. Slot:
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Preparation of Potassium Tris-oxalatoferrate(III) Trihydrate ($K_3[Fe(C_2O_4)_3] \cdot 3H_2O$)

The balanced molecular equation for the formation of potassium tris-oxalatoferrate(III) trihydrate, $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$ is as follows:



A student dissolves 8.01 g of iron(III) nitrate nonahydrate, $Fe(NO_3)_3 \cdot 9H_2O$ in 40 mL of hot water followed by 7.68 g of potassium hydrogen oxalate monohydrate, $KHC_2O_4 \cdot H_2O$. When the solution is cool, 10 mL of 6.0 mol L^{-1} KOH is added carefully, neutralizing the solution and allowing the tris-oxalatoferrate(III) ion to form. Ethanol is added to lower the solubility of the product in the solution and 6.72 grams of the product, potassium tris-oxalatoferrate(III) trihydrate, $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$, are recovered by suction filtration after cooling the solution on the bench.

1. Calculate molar mass and mole quantities of all reagents.
2. Identify the limiting reagent. Show product yield calculations for each reagent to justify your choice.
3. Calculate the theoretical yield of potassium tris-oxalatoferrate(III) trihydrate, $K_3[Fe(C_2O_4)_3] \cdot 3H_2O$.

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4. Determine the percentage yield of the $\text{K}_3[\text{Fe}(\text{C}_2\text{O}_4)_3] \cdot 3\text{H}_2\text{O}$.